



JALDRISHTI

GROUND SURFACE • 0.00 MBGL

A GROUNDWATER MONITORING AND PREDICTIVE ALERT SYSTEM

MINOR PROJECT

B.TECH • COMPUTER SCIENCE & ENGINEERING

INVERTIS UNIVERSITY

2026

INDIA RUNS ON WATER IT CANNOT SEE

63%

OF IRRIGATED AREA

Fed from groundwater, not surface canals or rivers.

85%

OF RURAL DRINKING WATER

Drawn from wells and borewells across the country.

1st

LARGEST USER, WORLDWIDE

India extracts more groundwater than any other nation.

The resource is invisible, so it is managed **reactively**. A district usually learns its water table has fallen only after wells begin to fail.

THE MEASUREMENTS EXIST. THEY JUST CAN'T BE READ.

The Central Ground Water Board monitors a national network of observation wells and publishes every reading, every year — as hundreds of pages of tabulated PDF.

- A PDF table cannot be **queried**, ranked, or joined to anything else.
- Nothing in the document **compares** one region against another.
- Nothing in it looks **forward** — there is no station-level forecast.
- The raw depth figure isn't comparable: 10 m is routine in one aquifer and alarming in the next.

FROM PUBLISHED PDF TO A DECISION SURFACE

LAYER 01

EXTRACTION

Batch table extraction from the source report, with generated station identifiers.

→ raw records

LAYER 02

PREPARATION

Cleaning, quality control, and a per-station historical baseline.

→ clean set + baselines

LAYER 03

ANALYTICS

The GAVI score, then a prioritised alert engine over score and trend.

→ scored + alerted

LAYER 04

FORECASTING

One- and three-year projections per station, converted back into scores.

→ forward alerts

LAYER 05

PRESENTATION

An interactive dashboard built for officials, not analysts.

→ live dashboard

Each layer persists its own dataset, so any stage can be re-run and refined without repeating the expensive ones upstream.

A DECADE OF READINGS, MADE QUERYABLE

86,912

OBSERVATIONS

9,545

MONITORING STATIONS

713

DISTRICTS

31

STATES & UTS

Covering **2015 to 2024**. January readings are used throughout, so every station is compared against itself at the same point in the annual recharge cycle — the seasonal swing is removed by construction, not by modelling.

GAVI: ONE NUMBER THAT TRAVELS

$$\text{GAVI} = 100 \times [1 - (\text{WL_current} - \text{WL_min}) \div (\text{WL_max} - \text{WL_min})]$$

Each station is rescaled against **its own historical extremes**. 100 is that station at its best on record; 0 is its worst.

Because the normalisation is station-specific, local geology and aquifer type are absorbed into the baseline — which is what makes a score of 30 in Rajasthan mean the same thing as a score of 30 in Odisha.

EVERY BAND NAMES A DECISION

■ SAFE	75 – 100	Good availability. Continue routine monitoring.
■ WATCH	50 – 74	Moderate stress developing. Increase observation frequency.
■ STRESSED	25 – 49	Significant depletion. Regulate demand and plan recharge.
■ CRITICAL	0 – 24	Severe deficit. Emergency intervention and extraction control.

A district officer reads a category and an action — never a hydrological unit.

ONE STATION, ONE ALERT, ONE YEAR

01	CRITICAL_GROUNDWATER	GAVI below 25
02	DEPLETION_WARNING	GAVI below 50 and the table is falling
03	SUDDEN_DROP	Level falls 2.0 m or more in one year
04	RECOVERY_SIGNAL	Level rises 1.0 m or more in one year
05	NORMAL	No rule satisfied; routine monitoring

The ladder is strictly ordered — severity always outranks trend, so no station can carry contradictory signals.

AN ALERT NOBODY BELIEVES IS WORTHLESS

A single bad reading — a faulty gauge, one anomalous season — must not put a district into emergency response. Two filters stand between the data and the siren:



2x

PERSISTENCE CHECK

A critical alert must appear in two or more consecutive observations before it is treated as confirmed.

3obs

ROLLING WINDOW

A three-observation window smooths transient spikes so measurement noise cannot trigger a response.

DELIBERATELY SIMPLE, ON PURPOSE

$$WL_future = WL_last + \text{mean historical } \Delta WL$$

Each station is projected at **one-year** and **three-year** horizons, clipped to the physically meaningful range of 0–50 mbgl, then converted back into a GAVI score against the same baseline.

Two reasons the first model is a transparent trend and not a neural network: it can be **explained to the official acting on it**, and it sets the benchmark any machine-learning model has to beat to justify its opacity.

FIT ON 2015–2022. PREDICT 2023. SCORE IT.

WITHIN 1 M



61.7%

WITHIN 2 M



80.7%

1.38m

MEAN ABSOLUTE ERROR

19.2%

MAE AS A SHARE OF MEAN LEVEL

Four stations in five land within two metres. That is honest baseline accuracy — reported as measured, and the number the next model must improve on.

WHAT THE DATA ALREADY SAYS

43.3%

OF STATIONS UNDER STRESS (GAVI BELOW 50)

1,109

CRITICAL ALERTS ACTIVE

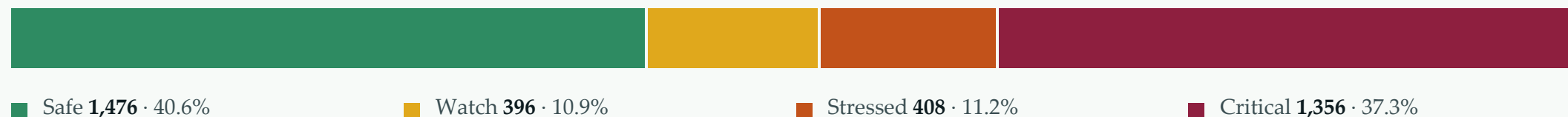
54.1

NATIONAL AVERAGE GAVI

Every one of these figures was already sitting in the published record. It was spread across hundreds of pages of tabulated PDF, where it is unreadable at a glance and unusable as a trigger for action.

AND WHERE IT IS HEADING

PROJECTED GAVI CATEGORY AT THE THREE-YEAR HORIZON — 3,636 STATIONS WITH SUFFICIENT RECORD



1,356 stations are projected to be critical within three years, against 1,263 flagged at the one-year horizon. Those are named stations in named districts — a work list, not a statistic.

SIX PAGES, ONE BROWSER TAB

OVERVIEW

National position at a glance — headline counts and the current distribution.

GAVI ANALYSIS

Score distributions and category movement over the decade.

ALERTS

Live alert counts by type, filtered down to state and district.

GEOGRAPHIC

Clustered station maps and district ranking by stress ratio.

FORECASTING

One- and three-year projections with forward-looking alerts.

STATION EXPLORER

The full history of any single well, against its own baseline.

No installation, no query language, no hydrology degree required.

OPEN SOURCE, END TO END

LANGUAGE	Python 3.11
EXTRACTION	Camelot · Ghostscript · pypdf · OpenCV
PROCESSING	pandas · NumPy · Jupyter
INTERFACE	Streamlit · Plotly · Folium
DEPLOYMENT	Docker · AWS EC2 · Git

Development cost is effectively nil: the toolchain is free, the data is published by the Government of India, and 86,912 records fit comfortably in memory on a commodity laptop. The only recurring cost is optional hosting.

SHORTEN THE DISTANCE BETWEEN MEASUREMENT AND INTERVENTION.

The readings were already being taken. JalDrishti turns them into a score that travels, an alert that can be trusted, and a forecast that arrives before the wells do.